Math 2263 Group Quiz 4

Names: _____

1. Let $P(x,y) = -y$ and $Q(x,y) = x$, let C be the curve that traces the unit circle once counterclockwise, and let R be the unit disk (the region enclosed by C). Calculate the work done by $\langle P, Q\rangle$ along C and then calculate $\iint_R (Q_x - P_y)\, dA$, verifying Green's theorem is correct in this example.

2. Evaluate the line integral $\int_C (x - y)\, ds$ where C is the line segment from (a,b) to (c,d).

3. Using a line integral, find the lateral surface area of the portion of the cylinder $x^2 + y^2 = r^2$ above the plane $z = 0$ and below the graph of $z = f(x, y)$. (Leave the answer as an integral involving f).

4. Let R be the set of points in the x, y-plane which satisfy $x^2 + y^2 \leq a^2$. Use the previous exercise to explain why

$$\iint_R f(x, y)\, dA = \int_0^a \int_0^{2\pi} f(r\cos\vartheta, r\sin\vartheta)\, r\, d\vartheta\, dr$$